Bisection vs Anderson–Björck:

A Comparison of Root-finding Methods

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Intro

These slides cover 4 root-finding methods:

1. Bisection
2. Regula falsi “method of false position”
3. Illinois method (Snyder, 1953; Dowell & Jarratt, 1971)
4. Anderson–Björck (Anderson & Björck, 1973)

I will assume you know the concept of bisection, so that will be a quick review/reference point for us

Motivation

Why even do this?-

Well, why even do anything? If you're going to do something, do it well.

More importantly, when it's 3:30am and you're trying to speed through a calibration, each iteration takes years off your life and decades off your rapidly dwindling sanity

- You'd like to shrink the initial brackets, but when targeting strong asymmetries (e.g. size distributions), things don't behave as nicely as you'd like
- Widening the brackets adds 1, maybe 2 iters, but you are no longer a sane person that properly allocates your time

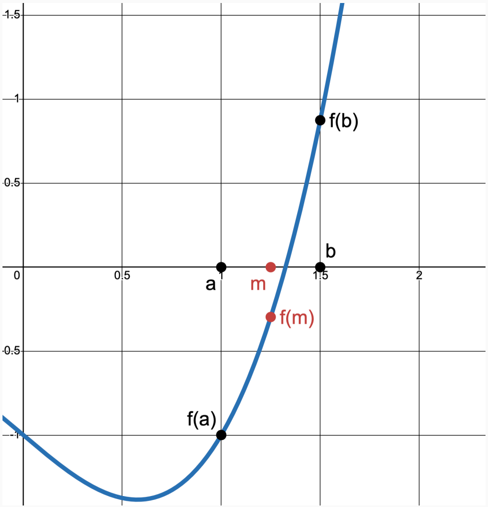
Bisection Review

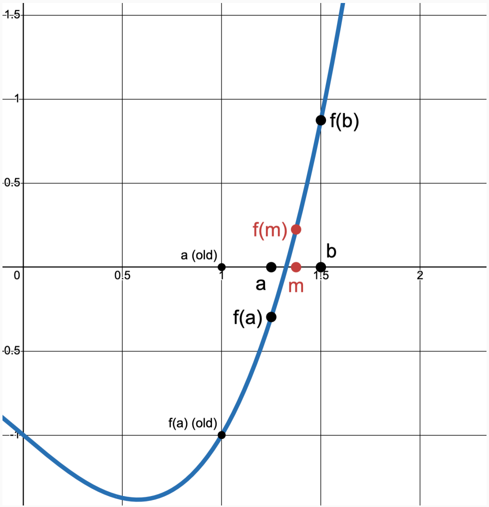
Suppose I wanted to find the root of $f(x)$, which is continuous between a and b:

1. Pick initial brackets (a and b) where $f(a)$ and $f(b)$ have opposite signs
2. Find the midpoint of a and b, $m = \frac{a+b}{2}$ and take $f(m)$
3. If $f(m)$ has the same sign as $f(a)$, replace a with m
    - Or, replace b with m if $f(m)$ has the same sign as $f(b)$
4. Continue until $|f(m)| < desired\_tol$. Then m sufficiently approximates the root

Next slide: example with $f(x) = x^3 - x - 1$ and initial $a = 1$ and $b = 1.5$

Bisection: Steps 1-3 (All graphs made with Desmos)





Bisection: Steps 4 - N

Now $m = 1.375$ becomes the new b and the process repeats until $f(m)$ is close to 0

- Determined by $|f(m)| < \epsilon$, where ϵ is sufficiently small and chosen by the modeler

This is a perfectly valid method, but we can make it faster with the “false position”

Regula Falsi

We use bisection, or various other numerical methods, because economics is hard

- What is simple is finding the root of a line

We can use the root of a line to better update our new a or b point:

1. Instead of the midpoint of a and b, draw a line between $f(a)$ and $f(b)$
2. Call the root of this line c and take $f(c)$
   - I'm using c since this is no longer a midpoint, but call it m if you like
3. If $f(c)$ has the same sign as $f(a)$, replace a with c
   - Or, replace b with c if $f(c)$ has the same sign as $f(b)$
4. Continue until $|f(c)| < desired\_tol$. Then c sufficiently approximates the root

Building the Line

To draw our line, linearly interpolate (and rearrange a bit) $f(a)$ and $f(b)$:

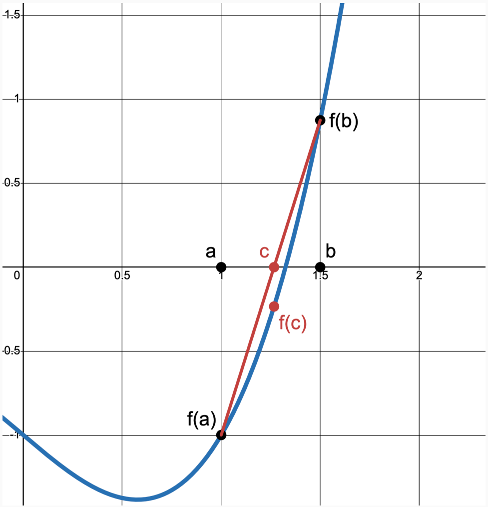
$$c = b - \frac{f(b) \times (b - a)}{f(b) - f(a)}$$

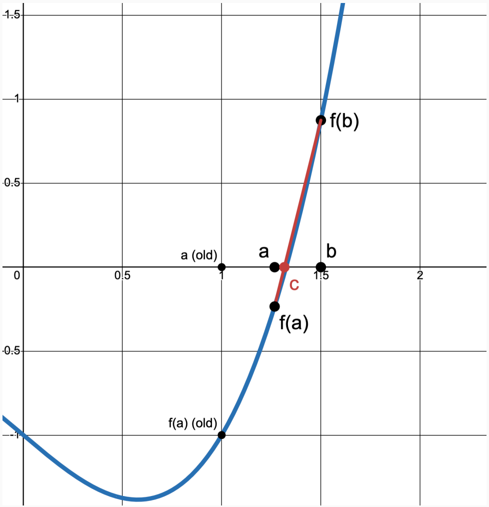
Optional: set a condition to use the smaller $|f(val)|$. If $|f(a)| < |f(b)|$, use:

$$c = \color{red}{a} - \frac{f(\color{red}{a}) \times (b - a)}{f(b) - f(a)}$$

- This likely wont matter

Regula Falsi: Steps 1-4

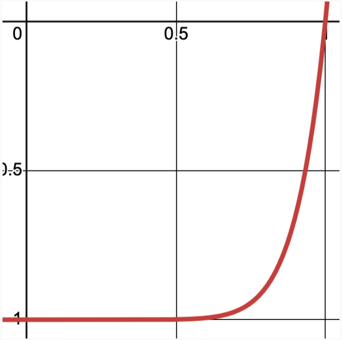




Issues?

Consider the case of $f(x) = x^{10} - 1$ on $[0, 1.3]$

- $f(x)$ is flat near 0 and blows up near 1.3
- Our line always crosses the axis around the left end
- $b = 1.3$ is retained forever and a crawls towards 1 (true root)



The Illinois Method

The Illinois method aims to correct this issue

- First as technical note by Snyder (1953), then formalized by Dowell & Jarratt (1971)

If sign $f(c)$ = sign $f(a)$, then the root is in $[c, b]$

- c is the new a; b remains the same

If this happens twice in a row, assign $f(b) = \frac{f(b)}{2}$ (same for $f(a)$ if opposite case)

- If it happens again on the next iter, halve again
- *Everything else remains the same as in regula falsi*
- Convergence guaranteed (halving does not change sign; stale bracket replaced)
  - ▷ Order of convergence ≈ 1.442 (Dowell & Jarratt, 1971)

Regula Falsi Failure and Illinois Correction for $f(x) = x^{10} - 1$

Regula Falsi:

```
iter=1   a=0.0   b=1.3   c=0.0942995953723274
iter=2   a=0.0942995953723274   b=1.3   c=0.18175887251907943
iter=3   a=0.18175887251907943   b=1.3   c=0.26287401252030435
iter=4   a=0.26287401252030435   b=1.3   c=0.3381051033222695
iter=5   a=0.3381051033222695   b=1.3   c=0.4078779165927524
iter=6   a=0.4078779165927524   b=1.3   c=0.47258315356239
iter=7   a=0.47258315356239   b=1.3   c=0.5325715106320142
iter=8   a=0.5325715106320142   b=1.3   c=0.5881445691706799
iter=9   a=0.5881445691706799   b=1.3   c=0.6395439707682159
iter=10   a=0.6395439707682159   b=1.3   c=0.6869431667412391
iter=11   a=0.6869431667412391   b=1.3   c=0.7304464366929381
iter=12   a=0.7304464366929381   b=1.3   c=0.7700987444968075
iter=13   a=0.7700987444968075   b=1.3   c=0.8059075090169241
iter=14   a=0.8059075090169241   b=1.3   c=0.8378738914244527
iter=15   a=0.8378738914244527   b=1.3   c=0.8660276320873561
iter=16   a=0.8660276320873561   b=1.3   c=0.8904572813509999
iter=17   a=0.8904572813509999   b=1.3   c=0.911328251308228
iter=18   a=0.911328251308228   b=1.3   c=0.9288846695992143
iter=19   a=0.9288846695992143   b=1.3   c=0.9434360517492835
iter=20   a=0.9434360517492835   b=1.3   c=0.9553339751921469
iter=21   a=0.9553339751921469   b=1.3   c=0.9649455723416893
iter=22   a=0.9649455723416893   b=1.3   c=0.9726296888054463
iter=23   a=0.9726296888054463   b=1.3   c=0.9787191336334624
```

Max iter hit at 100

Illinois:

```
iter=1   a=0.0   b=1.3   c=0.0942995953723274
iter=2   a=0.0942995953723274   b=1.3   c=0.18175887251907943
iter=3   a=0.18175887251907943   b=1.3   c=0.3330171567671213
iter=4   a=0.3330171567671213   b=1.3   c=0.5634423147022629
iter=5   a=0.5634423147022629   b=1.3   c=0.8463635731395354
iter=6   a=0.8463635731395354   b=1.3   c=1.07491017706849З
iter=7   a=0.8463635731395354   b=1.07491017706849З   c=0.9454923183277778
iter=8   a=0.9454923183277778   b=1.07491017706849З   c=0.9828011093189349
iter=9   a=0.9828011093189349   b=1.07491017706849З   c=1.0040954923602121
iter=10   a=0.9828011093189349   b=1.0040954923602121   c=0.9996755374546944
iter=11   a=0.9996755374546944   b=1.0040954923602121   c=0.9999940615677148
iter=12   a=0.9999940615677148   b=1.0040954923602121   c=1.000005704633256
iter=13   a=0.9999940615677148   b=1.000005704633256   c=0.9999999998475554
iter=14   a=0.9999999998475554   b=1.000005704633256   c=0.9999999999999961
```

The Anderson–Björck Correction

If sign $f(c) = f(a)$, instead of $f(b) = \frac{f(b)}{2}$, Anderson–Björck uses $k \times f(b)$, where:

$$k' = 1 - \frac{f(c)}{f(a)},$$

$$k = \begin{cases} m' & \text{if } m' > 0, \\ \frac{1}{2} & \text{otherwise.} \end{cases}$$

Dynamic adjustment speeds up the convergence based on the distance of y values

- Updating happens each time, not after two matching signs in a row
- Hasn't changed Illinois speed for me personally
  - Order of convergence ≈ 1.7 (Anderson & Björck, 1973)

My JMP D-StSt: Bisection vs Anderson–Björck

Bisection:

```
========================================================================
 METHOD 1: BISECTION    bracket [5, 10],  xtol = 1e-07 on the price
========================================================================
```

iter	pval	wage	C	g=C-1/p
1	5.0000000	0.428000	0.030432	-1.70e-01
2	10.0000000	0.214000	0.568592	+4.69e-01
3	7.5000000	0.285333	0.167265	+3.39e-02
4	6.2500000	0.342400	0.077432	-8.26e-02
5	6.8750000	0.311273	0.115721	-2.97e-02
6	7.1875000	0.297739	0.139706	+5.76e-04
7	7.0312500	0.304356	0.127275	-1.49e-02
8	7.1093750	0.301011	0.133378	-7.28e-03
9	7.1484375	0.299366	0.136514	-3.38e-03
10	7.1679688	0.298550	0.138103	-1.41e-03
11	7.1777344	0.298144	0.138903	-4.17e-04
12	7.1826172	0.297942	0.139304	+7.92e-05
13	7.1801758	0.298043	0.139104	-1.69e-04
14	7.1813965	0.297992	0.139204	-4.48e-05
15	7.1820068	0.297967	0.139254	+1.72e-05
16	7.1817017	0.297980	0.139229	-1.38e-05
17	7.1818542	0.297973	0.139241	+1.69e-06
18	7.1817780	0.297976	0.139235	-6.07e-06
19	7.1818161	0.297975	0.139238	-2.19e-06
20	7.1818352	0.297974	0.139240	-2.52e-07
21	7.1818447	0.297974	0.139241	+7.17e-07
22	7.1818399	0.297974	0.139240	+2.32e-07
23	7.1818376	0.297974	0.139240	-1.02e-08
24	7.1818388	0.297974	0.139240	+1.11e-07
25	7.1818382	0.297974	0.139240	+5.04e-08
26	7.1818379	0.297974	0.139240	+2.01e-08
27	7.1818377	0.297974	0.139240	+4.95e-09
28	7.1818376	0.297974	0.139240	-2.62e-09

```
 METHOD 1: BISECTION:  p = 7.1818376    28 solves total    161.22 seconds
```

Anderson–Björck:

```
========================================================================
 METHOD 3: ANDERSON-BJORK    bracket [5, 10],  xtol = 1e-07 on the price
========================================================================
```

iter	pval	wage	C	g=C-1/p
1	5.0000000	0.428000	0.030432	-1.70e-01
2	10.0000000	0.214000	0.568592	+4.69e-01
3	6.3285720	0.338149	0.081606	-7.64e-02
4	7.1688375	0.298514	0.138174	-1.32e-03
5	7.1835141	0.297904	0.139378	+1.70e-04
6	7.1818333	0.297974	0.139240	-4.41e-07
7	7.1818377	0.297974	0.139240	+1.85e-11
8	7.1818377	0.297974	0.139240	-1.67e-16

```
 METHOD 3: ANDERSON-BJORK:  p = 7.1818377    8 solves total    46.01 seconds
```

(This uses xtol ($|b - a| < \epsilon$), but it's the same principle)

Other Methods?

Many other weights exist, along with a multitude of completely different methods, which may be faster

HOWEVER, Anderson–Björck has three benefits in one: speed, [relative] ease of learning, and guaranteed to work (again, on a continuous on a sign-change bracket)

On Brent (1971), I think this is what ‘fzero’ does in MATLAB, which if you’re in this class, you’re NOT using anyway. Regardless, the decision tree you’ll have to follow for Brent is annoying and the path can move a lot with different warm starts. Not something your K-S regression coefficients will like

References

- Anderson, N. & Å. Björck (1973). **"A New High Order Method of Regula Falsi Type for Computing a Root of an Equation".** In: *BIT* 13.3, pp. 253–264. DOI: *10.1007/BF01951936*.
- Brent, R. P. (1971). **"An Algorithm with Guaranteed Convergence for Finding a Zero of a Function".** In: *The Computer Journal* 14.4, pp. 422–425. DOI: *10.1093/comjnl/14.4.422*.
- Dowell, Mark & Peter Jarratt (1971). **"A Modified Regula Falsi Method for Computing the Root of an Equation".** In: *BIT* 11.2, pp. 168–174. DOI: *10.1007/BF01934364*.
- Snyder, James N. (1953). ***Inverse interpolation, a real root of*** $f(x) = 0$. ILLIAC I Library Routine H1-71. University of Illinois Digital Computer Laboratory, p. 4.

Graphs: https://www.desmos.com/calculator